Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Plane shapes: types, properties and perimeter. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	2
Topic	Plane Shapes (2-D Shapes)
Subtopic	Plane shapes: types, properties and perimeter
Theme	—
Curriculum objectives	• identify the types of plane shapes and state their properties, similarities and differences • identify the parts of a circle • find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle
Previous knowledge	• Students have been taught angles.
Lesson sequence	• identify the types of plane shapes and state their properties, similarities and differences — the teacher draws the various shapes on the board; the student identify their properties; the teacher writes the properties of the shapes on the board for the students to learn • identify the parts of a circle — the teacher explains the parts of a circle to the students: the arc, radius, chord and diameter, with the circumference, segment, sector and tangent labelled on the board drawings • find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle — the teacher works the perimeter of plane shapes on the board: square, rectangle, trapezium, parallelogram, regular hexagon and circle, including finding a missing side or radius when the perimeter is given
Instructional materials	• Cardboard containing angles presentation.
Evaluation	• Find the perimeter of a square of sides 8cm • Find the perimeter of a rectangle of length 7cm and breadth 4cm. • What is the perimeter of regular pentagon with sides 5cm. • How many triangles can an octagon be divided into if you draw diagonals from one vertex to all non-adjacent vertices? • Find the perimeter of a parallelogram whose base is 6cm and sides 5cm • The circumference of a circle is 22cm – Find the radius of a circle ($\pi = \frac{22}{7}$).

	• A rectangle is such that its length is twice its breadth, if the breadth is d, find the total length of the sides of rectangle. • Calculate the perimeter of the figure below. [the figure is not included here, marked 1cm, 5cm, 9cm] • Find the perimeter of a regular hexagon with sides 5cm.

Lesson objectives

1.	identify the types of plane shapes and state their properties, similarities and differences
2.	identify the parts of a circle
3.	find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Plane shapes: types, properties and perimeter

Class: JSS1 **Subject:** Mathematics **Week:** 2

Section 1: Getting to Know Plane Shapes

When you look at a flat surface — a page of your book, a floor tile, the face of a clock — you are looking at a **plane shape**. A plane shape is a flat, two-dimensional (2-D) figure. It has length and width, but no thickness or height. We call these shapes **plane figures**, and studying them is all about noticing their edges (sides), their corners, and their angles.

There are two big families of plane shapes.

Polygons are closed plane shapes made only of straight sides. The name comes from *poly* (many) and *gon* (angles). Some common polygons are:

- **Triangle** — 3 sides and 3 corners (vertices).
- **Quadrilateral** — 4 sides and 4 corners. Squares, rectangles, trapeziums and parallelograms are all quadrilaterals.
- **Pentagon** — 5 sides.
- **Hexagon** — 6 sides.
- **Octagon** — 8 sides.

A polygon is called **regular** when all its sides are equal in length and all its angles are equal. For example, an equilateral triangle and a square are regular polygons, but a rectangle with unequal adjacent sides is not regular.

A **circle** is different. It is a plane shape, but it has **no straight sides and no corners**. It is the set of all points at the same distance from a fixed centre point.

A good way to compare plane shapes is to look at how many sides they have, whether the sides are straight or curved, whether they have corners, and whether all the sides are equal. Using these, you can tell shapes apart and spot what they have in common.

Let's zoom in on the parts of a circle, because they are useful for telling circle shapes apart from polygons.

- **Centre** — the fixed point at the very middle.
- **Radius** — the line from the centre to any point on the circle. It is the same length all the way around.
- **Diameter** — a straight line that passes through the centre and touches the circle at two points. The diameter is twice the radius.
- **Circumference** — the whole curved distance all the way around the circle.
- **Arc** — part of the circumference.
- **Chord** — a straight line joining two points on the circle (the diameter is the longest chord).
- **Sector** — a "slice" of the circle, like a slice of cake, bounded by two radii and an arc.

Now compare a circle with a square. A square has 4 equal straight sides and 4 right-angle corners; a circle has one continuous curved boundary, no corners, and every point on its edge is the same distance from the centre. Both are plane shapes, but they share no straight sides or corners. Noticing these differences is exactly how you describe the properties of any plane shape you meet.

Example 1 — Naming a shape and its properties. Look at a shape with 4 straight sides where only two of the sides are parallel. What is it, and how is it different from a square?

Solution. A four-sided plane shape is a quadrilateral. When exactly two sides are parallel and the other two are not, it is a **trapezium**. It differs from a square because a square has all four sides equal and all four angles right angles, whereas a trapezium has only one pair of parallel sides and its angles and side lengths are not all equal.

Example 2 — Describing a regular polygon. A shape has 5 equal straight sides and 5 equal angles. Name the shape and say which family it belongs to.

Solution. A plane shape with 5 sides is a **pentagon**. Because all 5 sides are equal and all 5 angles are equal, it is a **regular pentagon**. Since it is made of straight sides and is closed, it belongs to the family of **polygons** (it is also a closed figure with sides and corners, not a circle).

Example 3 — Comparing a circle and a parallelogram. State one property that a circle and a parallelogram share, and one way in which they are different.

Solution. **Similarity:** Both a circle and a parallelogram are plane (flat, 2-D) shapes. **Difference:** A parallelogram is a polygon made of four straight sides with two pairs of parallel sides and four corners, while a circle has one continuous curved boundary, no straight sides and no corners.

Practice

1. Name the plane shape with 4 straight sides where both pairs of opposite sides are parallel and all four sides are equal in length. Is it a regular polygon? Give a reason.
2. What is the name of the curved distance all the way around a circle? In a circle, how does the diameter compare to the radius?
3. A shape has 6 straight sides and 6 equal angles. Name the shape, and write down two properties of it.
4. State one property that a triangle and a pentagon have in common, and one way in which they are different.
5. Write down two properties that tell a parallelogram apart from a circle.

Section 2: Getting to Know the Circle

A circle is a flat, round shape. You meet circles everywhere: a coin, the face of a clock, the rim of a cup, the wheel of a bicycle. To describe a circle properly, and to talk about it the way mathematicians do, we give its different parts special names. Learning these names lets you say exactly which part of the circle you mean.

Every circle has one fixed point right at its middle, called the centre. It is the point that is equally far from every point on the circle's edge.

The circumference is the name for the boundary of the circle — the round outline that goes all the way around it. When we later find the distance around a circle, it is this circumference we are measuring.

Now for the lines that live inside a circle:

• A radius (plural: radii) is any straight line that joins the centre to the circumference. If you take any radius of the circle and any other radius, they are always the same length. That is why a circle is so perfectly round.

• A diameter is a straight line that passes through the centre and touches the circumference at both ends. The diameter is always exactly twice the length of a radius.

• A chord is any straight line joining two points on the circumference. A diameter is a special kind of chord that happens to pass through the centre — so it is the longest chord a circle can have.

Two more useful ideas:

• A semicircle is half of a circle. You get it by cutting along a diameter, so it is bounded by a diameter and half of the circumference.

• A quadrant is a quarter of a circle — half of a semicircle.

You will often see two other words about circles. An arc is part of the circumference itself, a curving piece of the round outline. A sector is a slice of the circle, shaped like a pizza slice: it is bounded by two radii and the arc between them. Keep these names in mind, because the centre, radius, diameter, chord, circumference, arc and sector are the parts of a circle you will keep meeting.

Worked example 1 — Naming the parts

A circle is drawn with its centre marked. A straight line from the centre to the edge is 4 cm long. Answer these questions about the circle.

(i) What is that line from the centre to the edge called?
(ii) What is the length of the diameter of this circle?

Solution
(i) A straight line that joins the centre of a circle to its circumference is a radius. Since every radius is the same length, any radius of this circle is 4 cm.
(ii) The diameter passes through the centre and touches the circumference at both ends, so it is twice the radius: Diameter = 2 × radius = 2 × 4 cm = 8 cm.

Worked example 2 — Deciding what each part is

On a circle, a straight line is drawn that joins two points on the circumference but does not pass through the centre. A second, curved piece of the round outline lies between the same two points.

(i) What is the straight line called?
(ii) What is the curved piece called?

Solution
(i) A straight line joining two points on the circumference is a chord. It is not a diameter, because it does not pass through the centre.
(ii) A curved piece of the circumference is an arc.

Practice

1. The radius of a circle is 6 cm. What is the length of its diameter?
2. State what each of the following parts of a circle is, and give a one-line reason for each answer:
   (i) the point equally far from every point on the circumference;
   (ii) a straight line joining the centre to the circumference;
   (iii) half of a circle.
3. A straight line joins two points on a circle's circumference and passes exactly through the centre. Is this line a chord, a radius, or a diameter? Explain your answer.

Section 3: Finding the Perimeter of Plane Shapes

The **perimeter** of a shape is the total distance all the way around its edge. You can picture it as the length of fence you would need to enclose the shape, or as the distance you would walk if you started at one corner and walked along every side until you got back to your starting point. Perimeter is a length, so it is always measured in units such as centimetres (cm), metres (m) or kilometres (km).

For most shapes you simply **add up the lengths of all the sides**. But a few shapes have special formulas that save you time, and one shape—the circle—has its own special name for its perimeter. Let us look at each one.

Square
All four sides of a square are equal. So if one side is of length s, the perimeter is

[\nP = 4 \times s\n]
which just means "four lots of one side".

Rectangle
A rectangle has two long sides (length l) and two short sides (width w). Opposite sides are equal, so the perimeter is

[\nP = 2 \times l + 2 \times w\n]
or, adding both the length and width and doubling,

\[\nP = 2 \times (l + w).\n\]
Regular polygon
A *regular* polygon has all sides equal and all angles equal. So if it has n equal sides, each of length s, its perimeter is simply

[\nP = n \times s.\n]
For example, a regular pentagon has 5 equal sides, so $P = 5 \times s$.

Trapezium
A trapezium has four sides, and usually they are all different. So you just add them all up:

[\nP = a + b + c + d\n]
where a, b, c, d are the four side lengths.

Parallelogram
A parallelogram has opposite sides equal. If the two pairs of equal sides have lengths a and b, then
[\nP = 2a + 2b = 2(a + b).\n]
This works just like a rectangle, because a rectangle is actually a special kind of parallelogram with right angles.

Circle
The perimeter of a circle has its own name: the **circumference**. If a circle has radius r, then
[\n\text{circumference} = 2\pi r.\n]
You may also see it written in terms of the diameter $d = 2r$, as πd. Use $\pi \approx 3.14$ unless you are told a different value, and round your answer to a sensible number of decimal places.

Notice the one shape where you must *not* just add sides: a circle has no straight sides, so its perimeter is found with the formula $2\pi r$ instead.

Let us work through each shape step by step so you can see exactly how the formulas are used.

Example 1: A square
A square has a side of length 5 cm. Find its perimeter.

Answer: $P = 4 \times s = 4 \times 5 = 20$ cm.

Example 2: A rectangle
A rectangle is 8 cm long and 3 cm wide. Find its perimeter.

Answer: $P = 2(l + w) = 2(8 + 3) = 2 \times 11 = 22$ cm.

Example 3: A regular polygon
A regular hexagon has six equal sides, each 4 m long. Find its perimeter.

Answer: A hexagon has $n = 6$ sides, so $P = n \times s = 6 \times 4 = 24$ m.

Example 4: A trapezium
A trapezium has sides 6 cm, 5 cm, 9 cm and 7 cm. Find its perimeter.

Answer: Add all four sides: $P = 6 + 5 + 9 + 7 = 27$ cm.

Example 5: A parallelogram
A parallelogram has one pair of sides 10 cm long and the other pair 6 cm long. Find its perimeter.

Answer: $P = 2(a + b) = 2(10 + 6) = 2 \times 16 = 32$ cm.

Example 6: A circle
A circle has a radius of 7 cm. Find its circumference, using $\pi \approx 3.14$.

Answer: circumference $= 2\pi r = 2 \times 3.14 \times 7 = 43.96$ cm.

Practice

1. A square has a side of length 6 cm. What is its perimeter?
2. A rectangle is 12 m long and 5 m wide. Find its perimeter.
3. A regular octagon has sides of length 3 cm each. What is its perimeter?
4. A trapezium has side lengths 8 cm, 6 cm, 10 cm and 7 cm. Find its perimeter.
5. A parallelogram has one pair of opposite sides 9 cm long and the other pair 4 cm long. What is its perimeter?
6. A circle has a radius of 10 cm. Find its circumference using $\pi \approx 3.14$.

Check your work

Attempt every question before you read this page.

Getting to Know Plane Shapes

1. A square. Yes, it is a regular polygon because all four sides are equal and all four angles are equal (each a right angle).

A four-sided plane shape is a quadrilateral. Both pairs of opposite sides parallel makes it a parallelogram, and all four sides equal makes it a rhombus — with all angles equal it is a square. A regular polygon needs equal sides and equal angles, which the square satisfies, so it is regular.

2. The curved distance all the way around a circle is its circumference. The diameter is twice the length of the radius.

   The circumference is the whole curved boundary of the circle. The radius runs from the centre to the edge, and the diameter passes through the centre touching two points on the circle, so diameter = 2 × radius.

3. A regular hexagon. It has 6 equal sides, 6 equal angles, and 6 corners (vertices); it is a closed polygon.

   Six sides means hexagon. Because all six sides are equal and all six angles are equal, it is a regular hexagon. Its properties: 6 straight sides, 6 equal angles, 6 vertices, and it is a closed plane figure.

4. Similarity: both are polygons made of straight sides with corners. Difference: a triangle has 3 sides and 3 corners, while a pentagon has 5 sides and 5 corners.

   Both are closed plane shapes built from straight sides, so both are polygons and both have vertices. They differ in number of sides: triangle = 3 sides, pentagon = 5 sides.

5. A parallelogram has 4 straight sides, two pairs of parallel sides, and 4 corners; a circle has one curved boundary, no straight sides and no corners.

   Any two of these work: parallelogram has straight sides while a circle is curved; parallelogram has four sides and four vertices while a circle has none; parallelogram has parallel opposite sides while a circle has no sides at all.

Getting to Know the Circle

1. 12 cm

   The diameter passes through the centre and is twice the radius: Diameter = 2 × 6 cm = 12 cm.

2. (i) centre — it is the fixed middle point; (ii) radius — a straight line from the centre to the circumference; (iii) semicircle — a circle cut in half by a diameter.

3. It is a diameter, and it is also a chord. A chord is any straight line joining two points on the circumference; a diameter is a special chord that passes through the centre, so it is the longest possible chord.

   The definition of a chord is any straight line joining two points on the circumference, so the line is a chord. Because it also passes through the centre, it is specifically a diameter — the longest chord a circle can have.

Finding the Perimeter of Plane Shapes

1. 24 cm

   A square has 4 equal sides. $P = 4 \times s = 4 \times 6 = 24$ cm.

2. 34 m

   $P = 2(l + w) = 2(12 + 5) = 2 \times 17 = 34$ m.

3. 24 cm

   An octagon has $n = 8$ sides. $P = n \times s = 8 \times 3 = 24$ cm.

4. 31 cm

   Add all four sides: $8 + 6 + 10 + 7 = 31$ cm.

5. 26 cm

   $P = 2(a + b) = 2(9 + 4) = 2 \times 13 = 26$ cm.

6. 62.8 cm

   circumference $= 2\pi r = 2 \times 3.14 \times 10 = 62.8$ cm.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-8b1f4cce	**Date**	
Subtopic	Plane shapes: types, properties and perimeter	**Rater**	

The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____